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SURVEY

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Applications of Distributed Ledger Technologies to the Internet of Things: A Survey

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Distributed Ledger Technologies (DLTs) and blockchain systems have received enormous academic, government, and commercial interest in recent years. This article surveys the integration of DLTs within another life-changing technology, the Internet of Things (IoT). IoT-based applications, such as smart home, smart transport, supply chain, smart healthcare, and smart energy, promise to boost the efficiency of existing infrastructures and change every facet of our daily life. This article looks into the challenges faced by such applications and reviews a comprehensive selection of existing DLT solutions to those challenges. We also identify issues for future research, including DLT security and scalability, multi-DLT applications, and survival of DLT in the post-quantum world.

CCS Concepts: • **Security and privacy → Security services**; **Distributed systems security**; **Privacy protections**;

Additional Key Words and Phrases: Distributed ledger technologies, blockchain, Internet of things

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1 INTRODUCTION

According to the formal definition established by International Telecommunication Union [1], the Internet of Things (IoT) refers to a world composed of networked physical objects, where everything potentially has a unique identifier and can interact and cooperate with each other for common goals. With the rapid development of information technologies and manufacturing of smart devices, IoT has become a critical technology to enable a wide variety of devices to communicate and share data with each other. It is anticipated that there will be over 20 billion devices, ranging

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from smartphones to wearables and simple sensors, connected to the Internet in the next 10 years [2]. Meanwhile, the market value of IoT is expected to increase up to 724.2 billion U.S. dollars by 2023 [3], mainly fueled by the increasingly ubiquitous smart terminals and the mass data gathered by them.

Without any doubt, IoT applications are changing quietly every facet of our daily life. At home, intelligent inter-connected devices that can be remotely accessed and monitored provide family members with comfort and security. In the outside, major industry and society sectors, such as supply chain, healthcare, energy, and transports, are boosting efficiency by adopting IoT technology. However, some serious challenging issues still need to be tackled before the advent of the IoT era [4–6].

The aim of this article is to survey how a range of Distributed Ledger Technologies (DLTs), emerging in recent times, have been applied to tackle issues with the IoT and to explore potential applications of DLTs to IoT. A distributed ledger is not maintained by any central authority and can spread across a peer-to-peer network. Each network participant can record, share, and synchronize transactions in the ledger. DLT has made it possible to quickly deploy transparent, distributed, and secure systems that run in a decentralized ledger. Transparency follows from the fact that such ledgers are typically public and openly shared among users, who are given the power to verify and accept/reject actions of the system (called transactions). An important property of a decentralized ledger is immutability, which means that once a transaction is accepted and added to the ledger by the majority of users, then it is computationally hard to revert the system back to a previous state. This is achieved via a combination of cryptographic puzzles, incentive mechanisms, and authentication protocols that altogether keep the ledger secure.

This survey aims to address the following question: How is DLT being used in the IoT and what potential issues may DLT solve? For example, IoT systems can collect users' personal data, including location and medical records. In this context, a decentralized secure network realized by a DLT has the potential to support a secure, private, and trusted IoT universe [7]. Other challenges in the IoT are identity management and traceability. Existing solutions to these issues, in particular those dedicated to meet the requirements of the traditional Internet, cannot be deployed in IoT due to the resource constraint and huge scale of IoT devices. Hence, new approaches and technologies for meeting the requirements of IoT domains are needed to address these challenges.

DLT provides a means to facilitate trusted interactions among IoT devices. It is a response to the current call for greater machine-to-machine or thing-to-thing communications, where encounters between devices are sparse and often unpredictable. Important properties of distributed ledgers that make them such a promising tool for the future of IoT are the following:

- *Decentralization*: IoT devices may act as autonomous agents. Even if the IoT ecosystem (network) is split due to some network link changes, IoT devices should be capable of correctly operating. Centralized systems can only partially achieve this by introducing recovery or re-synchronization mechanisms. The need for decentralization in IoT connects with the notion of the decentralized ledger architecture, which we will describe later. Properties of the IoT ecosystem can be ensured without relying on any trusted central parties. An advantage of DLTs is the decentralization away from a central authority—which is particularly attractive in IoT settings where no single authority can (or should) manage, hold, or own all data or be trusted to do so. DLTs provide an alternative away from a single point of attack or failure, as well as enabling protection from tampering by any single party.
- *Immutability:* Actions of devices and interactions among things in the IoT can be audited and traced via a ledger.
- *Distributed:* Distribution and sharing of the system state, in combination with the auditing and assurance property above, improves transparency throughout the IoT lifecycle.

In a nutshell, a technology that is called to be heavily involved in people's everyday activities, that is, the IoT, ought to be secure, transparent, robust, auditable, and reliable. DLT seems to offer these properties. And we will show in this article how IoT applications (will) benefit from it.

1.1 Contributions

Distributed ledger technologies have gained great attention in recent years. At the same time, various survey papers have been published exploring fundamental problems in blockchain and DLTs. For example, the authors in Reference [8] take a detailed technical look at the Bitcoin protocol and point out other potential applications besides cryptocurrency. The surveys [9] and [10] focus on security and privacy issues in Bitcoin and Bitcoin-like systems, whereas Reference [11] explores how blockchain can solve fundamental security challenges of traditional centralized systems. Meanwhile, there are several surveys focusing on the state of the art in the area of the integration of IoT with DLTs [12–19, 96]. In Reference [12], the authors explore 18 existing use cases of blockchain in which only four cases are explicitly designed for IoT. In Reference [13], the authors outline some state-of-the-art blockchain solutions that meet the security requirements of IoT—the review mainly focused on reviewing and categorizing the security issues according to a IoT hierarchical architecture. In Reference [14], Yeow et al. focus on the decentralized consensus mechanisms in IoT systems, especially the edge-centric IoT scenario. In Reference [15], the authors also focus on securing IoT using blockchain technology and also explore some blockchain-based IoT applications, such as smart home, electrical transactions. A brief overview of blockchain-based IoT and an exploration of how to design blockchain systems for IoT is given in Reference [16]. In Reference [17], the authors look into challenges and benefits of the combination of IoT and blockchain from a security point of view. They also describe a few blockchain platforms related to IoT. In Reference [18], the authors explore how blockchain can be used to overcome the inherent shortcomings (particularly in terms of privacy and security) of traditional IoT architecture. Moreover, there are some surveys focusing on the application of blockchain in some specific aspects of IoT, such as smart industry [19].

To summarize, there exists already a number of survey papers connecting IoT and DLT. They mostly focus on security and privacy issues, while neglecting other key areas in the IoT, such as identity management and machine-to-machine transactions. We will cover this gap. We will also expand the coverage of IoT application scenarios utilizing DLTs, such as supply chain and smart transport. A comparison table of the different aspects (un)described by previous surveys and those the reader will find in this article is given in Table 1. The highlights of such comparison is provided in the following list, which we use as our list of contributions.

(1) Compared to the existing survey papers, this article provides a more in-depth summary of how DLTs can help with different requirements of IoT.
(2) This article presents a comprehensive survey of different DLT-IoT combined applications, ranging from smart home, smart transport, supply chain, and smart healthcare to smart energy.
(3) We systematically present and discuss a comprehensive selection of existing DLT-based solutions for IoT. Both the advantages and limitations of these solutions are discussed. We go beyond the typical blockchain ledgers to consider variations in the area, such as directed acyclic graph structures, instead of the standard blockchain.
(4) We provide an overview of existing DLTs platforms that have the potential to support privacy and traceability in IoT applications. Strengths and weaknesses of these platforms are scrutinized and compared.
(5) Finally, we identify a number of open research directions toward the successful integration of DLT and IoT.

Table 1. Comparison with Existing Surveys

Ref.	How DLTs Benefit IoT				IoT scenarios using DLTs					Distributed Ledger Platfroms for IoT
	Security and Privacy	Identity management	Machine-to-Machine Transactions	Traceability and Provenance	Smart Home	Smart Transport	Supply Chain	Smart Healthcare	Smart Energy	
[12]	✓	×	×	×	×	×	×	×	×	×
[13]	✓	✓	×	×	×	×	×	×	×	×
[14]	×	×	×	×	×	×	×	×	×	✓
[15]	✓	×	✓	×	✓	×	×	×	×	✓
[16]	✓	×	✓	×	✓	✓	✓	✓	✓	×
[17]	✓	×	×	×	×	×	×	×	×	✓
[18]	✓	×	×	×	✓	×	×	×	✓	×
[19]	✓	×	×	×	×	×	×	×	×	×
Ours	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

1.2 Organization

The remainder of this survey is structured as follows. Section 2 will present some theoretical basics of DLTs. Section 3 describes how and where to apply the distributed ledger technologies in IoT. Section 4 is a comparison of existing DLTs that are already being used or show potential to have a positive impact in IoT. Section 5 presents a discussion on a few compelling open research issues. Section 6 concludes the article by indicating future trends in this field and research directions.

2 DISTRIBUTED LEDGER TECHNOLOGY: AN OVERVIEW

Digital ledgers, consisting of time-stamped digital records, have been at the heart of electronic services since the irruption of computers, replacing traditional ledgers used to record money and property. Now a mix of cryptographic puzzles, incentive mechanisms, and security protocols is enabling the collaborative creation of digital distributed ledgers [20], which seems to have no precedence in ancient human history. This boosts trust and transparency and opens the door to innovative applications that no longer rely on a centralized architecture.

This section presents a preliminary overview of Distributed Ledger Technology (DLT), which provides the background for the further discussions on the potential of DLT in IoT scenarios. As shown in Figure 1, we structure the section by describing how data are stored (Data layer), how users are connected (Network layer), how users agree on a ledger and resolve inconsistencies (Consensus layer), how a distributed ledger can become an execution environment for computer programs (Contract layer), and what type of applications can be built on top of DLT (Application layer).

2.1 Data Layer

The elementary unit of a distributed ledger is typically called a *transaction*. A transaction has a source (the sender), a signature that authenticates the source, and content (data). Transactions are gathered into blocks, and blocks are chained to each other creating a graph without cycle. Chained blocks free of cycle are needed to ensure that blocks can be partially ordered, making it easy to determine the evolution of the ledger.

Unsurprisingly, the most popular distributed ledger nowadays, the *Blockchain*, mimics traditional paper-based ledger, i.e., blocks are stored one after the other as they are added to the ledger. Widespread use of blockchain started with the introduction of the crypto-currency *Bitcoin* in 2009 [21]. Perhaps boosted by the financial crisis in 2008, Bitcoin offered an attractive decentralized

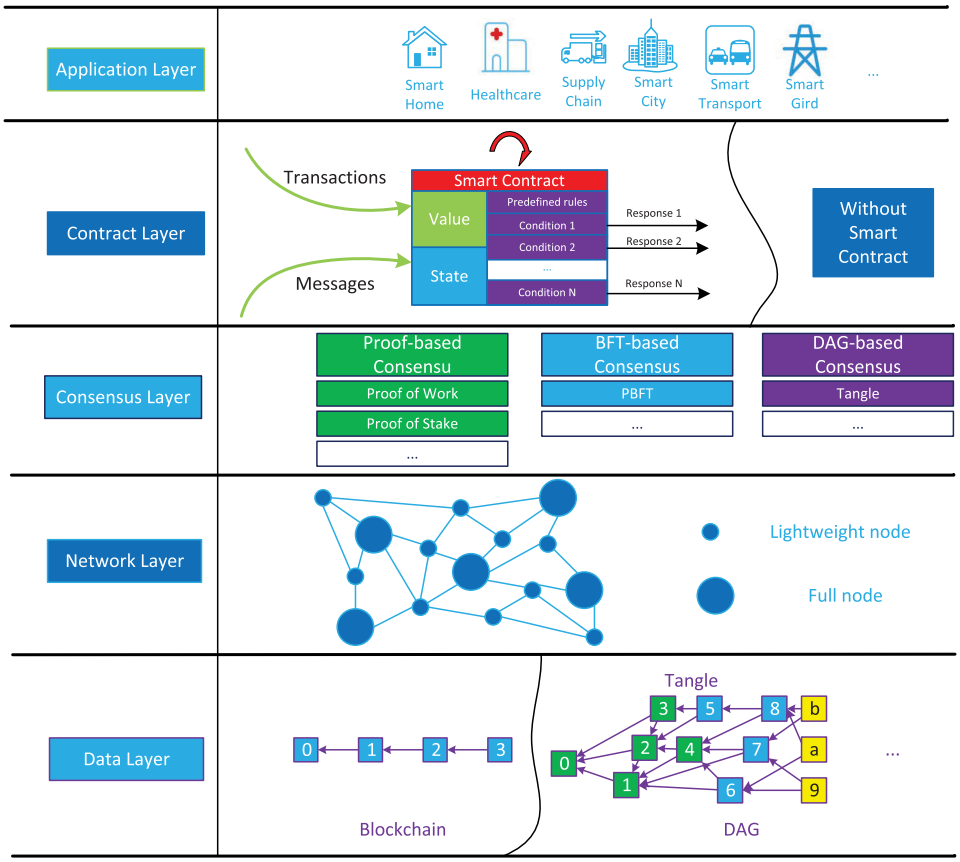

Fig. 1. The hierarchical architecture of a typical distributed ledger system.

solution to online monetary transactions. In a fully decentralize architecture, Bitcoin allows users to exchange digital cash in a consistent and secure way.

The logic of Bitcoin is simple. Users create transactions to send and receive coins (digital cash), and users themselves validate and append transactions to the blockchain. Fraudulent behavior is eliminated by making it computationally hard to add spurious transactions to the blockchain, allowing users to verify transactions, and incentivizing stakeholders to keep the blockchain growing.

From a technical viewpoint, each block in Bitcoin is linked to the previous block by embedding the hash of the previous block's header, called the pre-hash (see Figure 2). The pre-hash has the peculiarity that it must start with a series of zeros, making it more difficult to link two blocks as the number of zeros increases. The process of finding the correct pre-hash is called *mining* in Bitcoin, establishing a parallelism with gold mining as there is always a reward for the user that distributes a correct pre-hash. Hashes of transactions within a block are computed as well and stored in a Merkle tree, which guarantees the integrity of transactions in the block. Other metadata, namely *difficulty*, *timestamp*, and *nonce*, are used for the mining process in Bitcoin.

As the mode of adding blocks to a blockchain is sequential, it is unsuitable for systems requiring numerous transactions to be processed quickly. Another drawback of putting blocks in sequence is that the mining process becomes highly inefficient; a block is mined by many, added by just one. As a result, blockchain-based systems, such as Bitcoin, waste a disproportionate amount of energy and are considered by many as an environmental problem.

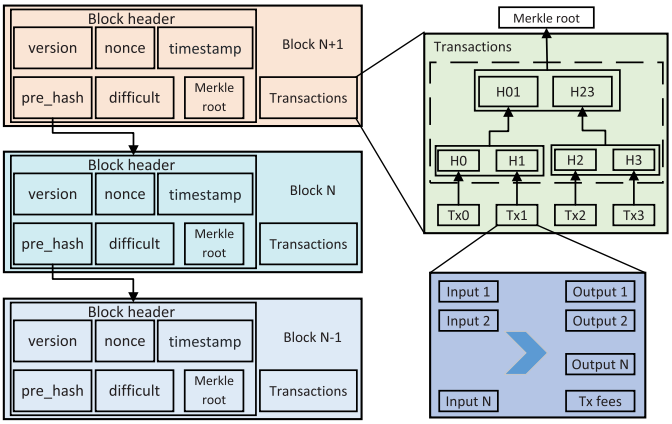

Fig. 2. The simplified block structure of Bitcoin.

A recent solution to the scalability and efficiency problem of blockchain is via generalizing the blockchain (a list) to a directed acyclic graph (DAG). Storing blocks or transactions in a DAG allows for parallelism of the mining process, given that there exist various nodes a new node can be linked to without creating a cycle. Popular systems using a DAG structure, such as IOTA [22], go further in terms of scalability and efficiency by eliminating third-party miners and letting transactions owners themselves to mine their transactions [24]. As a result, DAG-based systems have a comparative advantage in performance and scalability over blockchain. In terms of security, however, it has been shown that fewer resources are needed to launch a double-spending attack on DAG-based systems, which succeeds by owning 51% hashing power of the whole network in Bitcoin [21] and only 34% in IOTA [22]. Moreover, the security of current DAG platforms relies on central entities, called coordinators in IOTA [22] and witnesses in Byteball [23], which may lead to key points of failure.

2.2 Network Layer

The network architecture used by DLTs is peer-to-peer (P2P). Such a choice follows from the decentralization requirement, which is a core characteristic of DLT, and the fact that decentralization can only be achieved and maintained by a network architecture where participants/peers share resources without central entities [25]. Although all participants in a P2P network are considered equal to function as a service provider or requestor, they could be divided into different categories depending on their supporting functions, such as routing, database, wallet, and miner.

In the case of Bitcoin, every participant is responsible for the transmission and validation of transactions and blocks between nodes in the P2P network [26]. Other functions are optional, though. For example, full nodes keep a complete copy of the ledger, which allows them to autonomously verify the content of the transactions, such as double-spending. Lightweight nodes, e.g., wallets, only store the headers of all blocks, making it possible to verify consistency of the chain as it grows.

2.3 Consensus Layer

Consensus is the underlying mechanism used to ensure that each participant can have the correct copy of the ledger. Even when all participants are honest and only correct transactions are broadcasted and validated, the local view two agents have of the system state may differ due to

network latency or fault. Also, the risk of desynchronization or lack of agreement on the system state increases with the number of dishonest participants.

To this day, many consensus algorithms have been proposed and applied in different kinds of distributed ledger systems. Most of the existing consensus algorithms fall into the following three categories: proof-based, Byzantine Fault Tolerance–based (BFT-based), and DAG-based consensus algorithms.

Proof-based consensus. Accepting correct transactions ensures that a node has a consistent and correct local ledger, but it gives no guarantee on whether the node's local ledger is the same as the ledgers of other participants in the network. To reach decentralized agreement or consensus, a transaction(s) can be extended with a *proof* that ensures that the majority of nodes in the network will add such transaction(s) to their local ledger.

In Bitcoin, a proof of agreement is the *quickest* solution to a hard computational problem that links the current block (list of transactions) with the ledger, hence the name *proof-of-work*. The assumption is that nodes are programmed to accept the first correct incoming proof, rendering future proofs of the same type incorrect. The incentive for nodes to participate in such energy-hungry race[1] are transaction fees, which are given to the node that convinces the majority of the network to accept its proof.

In Ethereum, generating a proof of agreement is slightly more elaborated as, unlike in Bitcoin, not every node can work toward obtaining a correct proof. To prevent the racing problem of proof-of-work, Ethereum chooses a single validator through a probabilistic procedure that is biased toward nodes able to commit a high deposit (in virtual currency) to the network, hence the name proof-of-stake [27]. This type of consensus mechanism appeals to the rational behaviour of nodes in the network, that is, nodes are willing to neither risk losing their own deposits by misbehaving nor upset the stability of a system where they already hold a privilege position.

Although proof-of-work consensus is energy demanding, e.g., the energy consumed by the entire Bitcoin network surpasses that of many countries [29], it remains to be proven whether proof-of-stake can be as reliable as proof-of-work consensus mechanisms or simply reliably enough. In fact, proof-of-stake is known to suffer from a theoretical flaw known as the nothing-at-stake problem [30]. Forks, which occur when two proofs are equally valid and nodes in the network cannot determine who is the winner, are quickly resolved in a proof-of-work mechanism, because miners waste twice as much energy working on two ledgers than on one. As a consequence, miners are driven to focus on the longest ledger and forks are quickly disambiguated. Validators in a proof-of-stake mechanism are, in principle, not penalized for working on more than one fork, making it theoretically possible to have a large number of forks that will never be resolved.

PBFT-based consensus. In networks of limited size, consensus can be achieved by obtaining the state of each node in the network via a multiparty communication protocol. For a given transaction or block, all nodes can voice their agreement or disagreement with adding that block to their own ledger. Then, as usual, a majority rule is followed to determine whether to accept or reject the block. The earliest formalization of this intuition is by Castro and Liskov [33], who introduced Practical Byzantine Fault Tolerance (PBFT) to accomplish consensus provided that at most one third of network nodes are simultaneously faulty.

Various blockchain platforms, such as R3 [36], Stellar [37], BigChainDB [38], and IBM-backed Hyperledger [34], use a PBFT-based consensus algorithm. In Hyperledger, for example, whenever a new block needs to be created, a leader is voted to put forward such task. The leader is in charge of grouping transactions into a block, verifying its correctness, and broadcasting the created block to all validators. Upon reception, each validator checks and executes (in case of smart contracts)

[1]Other resources can be used to establish the proof-of-work race, such as bandwidth and storage [8].

Table 2. Comparison of Three Types of Consensus Algorithms

	Proof-based	**BFT-based**	**DAG-based**
Data structure	Mostly blockchain	Mostly blockchain	DAG
Number of validators	Mostly unlimited	Limited	Mostly unlimited
Type of DLs	Mostly permissionless DLs	Mostly permissioned DLs	Mostly permissionless DLs
Decentralization	High	Low	It depends
Networks scalability	High	Most low	High
Transaction throughput	Low	High	Very High
Forking	Yes	No	Yes
Confirmation speed of transactions	Low	Fast	Low
Transaction fee	Low	Mostly no	Very low or no

the received block. If correct, then the validator broadcasts its agreement. A block is finally added to the local ledger of a node if a given minimum number of validators agree with it, e.g., more than two thirds of the validators.

The threshold, say, Δ, used to decide whether to add or not a block to a local ledger is relevant. It is proven in Reference [33] that a PBFT-based consensus system is safe if the number of faulty validators f is lower or equal to one third of the total number of validators. This gives $\Delta = 2f + 1$ to be the lowest value of Δ that renders the system safe. The third-largest cryptocurrency, Ripple [35], for example, sets $\Delta = 5f + 1$ to improve reliability at the expense of complexity.

DAG-based consensus. Platforms, such as Byteball [23], Hashgraph [39], Dagcoin [40], and IOTA [22], distribute a ledger in the form of a DAG rather than a chain. To illustrate how consensus can be achieved on a DAG distributed ledger we take the tangle in IOTA as an example [22]. A node in the tangle is a transaction, instead of a block. To issue a new transaction (the yellow boxes in Figure 1), a node must randomly approve two other unconfirmed transactions (the blue boxes in Figure 1) by doing a lightweight proof-of-work. In this way, a subset of the transactions in the tangle are validated directly or indirectly by new transactions. With more transactions being added to the tangle, an early transaction can be validated by many new transactions. Then the confidence of a transaction is defined as the fraction of the number of new transactions referring to it in all new transactions. As shown in Figure 1, all green transactions are confirmed with high confidence.

Because adding a transaction to a DAG ledger is easy, DAG-based consensus is vulnerable to malicious participants adding and validating their own transactions, as in the PBFT-based consensus. The problem is aggravated by the fact that, unlike a blockchain, a DAG ledger allows redundant and inconsistent information to co-exist, while valid transactions are supposed to be found within the *heaviest* branch, in terms of confidence value, of the DAG ledger. This makes it hard for users to determine whether their transaction have been validated. The current workaround to this problem, at the expense of loosing decentralization, consists of a trusted validator (hard-coded in the system) with the power of raising the confirmation confidence of transactions to 100%, provided that they are correct.

Table 2 shows a comparison between three classes of consensus algorithms. First, both proof-based and BFT-based consensus algorithms are mainly applied in blockchain systems, whereas the underlying data structure of the DAG-based algorithms is a distributed DAG. Second, in BFT-based consensus, only limited permissioned validators can join the verifying network. Therefore, BFT-based consensus is adapted by permissioned distributed ledger systems, and the decentralization degree is lower than other consensus methods. There are also some central entities, such

as the coordinators used for the double checking of all transactions in IOTA. However, the IOTA foundation explains that the coordinator will be switched off in its mature stage. Furthermore, the throughput of DAG-based consensus is significantly higher than other two types of consensus, because it allows the appendation of multiple transactions simultaneously. Moreover, there is no fork in BFT-based consensus, and new transactions in a block are confirmed once they are appended to only the main blockchain.

In most permissionless ledger systems with proof-based consensus, the transaction fee is given to participants to incentivize them to validate transactions [28]. As the validators in BFT-based consensus are companies, banks, Internet service providers, or other organizations, such incentive becomes unnecessary. Furthermore, there is no transaction fee in some permissionless DAG ledger systems, where transactions are validated by new coming transactions, such as IOTA and Byteball.

Finally, we note that there are also some combinations of two or more different classes of consensuses. For example, the consensus adapted by Casper is a mixture of PoS and BFT [41], and both the PoS and BFT are used in the Mixin Network, which is a DAG-based platform [42].

2.4 Contract Layer

Smart contracts, first proposed by Szabo [43], are defined as computer protocols that contain a set of rules and are enforced to execute automatically. As a decentralized, transparent, tamper-resistant, and secure platform, the distributed ledger system is naturally the fertile ground for smart contracts. As implemented on distributed virtual machines provided by the underlying distributed ledger, each smart contract must be verifiable and can be auto-executed independently by every node in the network, e.g., in Ethereum [44].

As shown in Figure 1, every smart contract has its own assets and states with a unique address in the distributed ledger [44]. It contains all potential state transition rules and the corresponding actions. A contract can be triggered by transactions or messages from users or other contracts. The execution of the contract is determined by the predefined transition rules, which will change the state and assets of the contract.

By introducing Turing-complete languages to DLTs, any polynomial computational task can be executed by a smart contract [44]. Moreover, by using smart contracts for expressing business logic, many complex applications can be built on distributed ledgers. In other words, smart contracts makes the platforms of DLTs useful not only for cryptocurrencies but also for real-world applications.

2.5 Application Layer

Beyond cryptocurrencies, the application layer, which is the user-facing part of this architecture, can package a variety of decentralized solutions for specific scenarios. It allows third-party developers to build decentralized applications on top of underlying DLTs through the built-in interfaces. Although DLT is still immature, it clearly has the potential to empower applications with features, such as transparency and decentralization, that were either unfeasible before or too expensive to implement [45]. We end this section by highlighting some of those features.

- **Distributed and Decentralized.** Being distributed is a fundamental characteristic of DLTs. The distributed ledger is an autonomic system maintained by all participants through a well-designed consensus mechanism, where no central entity is needed.
- **Tamper-proof.** The new block confirmed by the consensus mechanism is added into the distributed ledger in an append-only way through cryptographic hash functions. As a result, both the consensus mechanism and hash functions (hashed on historical blocks) ensure that the historical records in the distributed ledger is almost impossible to be tampered with.

- **Transparency.** Each participant can keep a whole copy of the distributed ledger, which means all records in the ledger are open to tracing.
- **Automation and Programmability.** Smart contracts provide a programmatic interface to the underlying distributed ledgers and can be executed automatically.
- **Pseudo-anonymity.** In distributed ledger system, the pseudonym, known as the *address*, is used to conceal the real identity of each participant.

3 THE IMPACT OF DLTS ON THE INTERNET OF THINGS

In this section, we will explore how distributed ledger technologies benefit IoT in different aspects.

3.1 DLTs for IoT Security and Privacy

Due to the resource-constraint and widescale nature of the IoT, achieving standard security goals of networked systems, such as data confidentiality, integrity, authentication, access control, privacy, and trust, becomes a hard challenge [46, 47]. With the inherent cryptographic security and the nature of decentralization and trustlessness, DLT has been considered as a potential solution to the security challenges in the IoT. Next, we explore how DLTs are being used to cope with security and privacy issues in the IoT.

3.1.1 Data Confidentiality, Integrity, and Authentication. The IoT enables data to be shared among things and users for supporting decision-making. In such a data-sharing environment, data confidentiality, integrity, and authentication are fundamental security requirements. Confidentiality ensures that data are not leaked to unauthorized parties, integrity that data are not modified while in transit or at rest, and authentication that protocol participants pose the right credentials. The internet currently relies on a Public Key Infrastructure (PKI) to meet those three security goals, and the IoT is moving on a similar direction [49].

Two main types of PKI approaches exist, the certificate authority– (CA) based PKI and the web of trust– (WoT) based PKI. In CA-based PKI, the CA acts as a trusted third party issuing digital certificates that bind public keys to users [48]. The WoT-based PKI does not require any centralized certification authority. Instead, the signature of a given public key will be trusted if it is validated by other trusted users through a decentralized trust model.

While CA-based PKI has held the internet secure for a number of years, it is regarded as insufficient to cope with the ubiquitousness, mobility, and heterogeneity of IoT networks [50]. WoT-based PKI, on the other hand, cannot help to quickly establish trust between new devices joining the system [48]. In this context, distributed ledger technologies can be applied to build a distributed and secure PKI for the IoT [49]. As DLTs-based PKI are distributed, there is no single point of failure. Furthermore, in distributed ledger systems, trust is built by the consensus mechanism, any participant can join or exit the system at any time, and prior trustworthiness is not necessary.

A recent blockchain-based PKI has been proposed by Pinto et al. [50]. They use a web platform providing identity verification service, called Keybase, based on Pretty Good Privacy technology. Each Keybase user may have different accounts associated. In the verification process, on receiving a request from an unknown device B, device A should look up the records referring to device B in the blockchain. From the records, the user associated with device B is obtained. The public key linked to the user can be found in Keybase. This allows device A to verify the block signature and decide whether to trust device B.

DLTs can provide not only a distributed PKI for IoT but also additional integrity services with its inherent immutability (i.e., tamper-proof). In Reference [51], Liu et al. provide integrity services for IoT systems executed by smart contracts in Ethereum. Integrity is supported by the records in the blockchain that contain the hash values of data stored in the cloud.

3.1.2 Access Control. Access control refers to the selective restriction of access to resources. Existing access control mechanisms typically rely on a centralized architecture, in which a single point of failure has the potential to obstructs the operation of the whole system. As in the case of PKIs, DLTs can overcome the single-point-of-failure issue by providing trust in a decentralized way [52]. An implementation example is given in References [53, 54], where access control policies are stored in a blockchain. That said, how to adapt those approaches to the IoT ecosystem remains an open problem.

In References [55, 56], Ouaddah et al. proposed a blockchain-based access control framework (FairAccess). In their framework, all access control policies are encapsulated in transactions of a Bitcoin-like blockchain. Authorization is granted through a number of transactions types, such as GrantAccess, RequestAccess, DelegateAccess, and RevokeAccess. Instead of an exchange of cryptocurrency, Ouaddah et al. introduce the exchange of authorization tokens, allowing for secure exchange of access rights. However, due to the limited throughput performance and block size in Bitcoin, the scalability and granularity are major problems in this mechanism. Moreover, it seems difficult to set up a lightweight wallet in resource-constrained IoT devices. A follow-up work [57] suggests that implementing their framework in an Ethereum-like blockchain may improve scalability of the FairAccess system, whereas others [62] propose the use of various blockchains to handle different processes.

The overall improvement in performance shown by Ethereum over Bitcoin has made Ethereum the platform of choice for many blockchain-based access control implementations dedicated to the IoT. This translates to access control policies being embedded in Ethereum smart contracts. In Reference [58], for example, Zhang et al. use three types of smart contracts: The access control contract contains security policies, the judge contract is responsible for the judgment of illegal access, and the register contract is used to control and manage other contracts. Rifi et al. [59] use three types of smart contracts, too, in their architecture for securing IoT data access, while Ancile [60] considers that IoT applications managing health record require a more fine-grained division of responsibilities and propose the use of six smart contract types. Last, considering the difficulty of IoT devices acting as a node in blockchain systems, Novo [61] connects a set of IoT devices to a single node in the blockchain in such a way that trust can be established on a group of devices rather than on single device.

3.1.3 Privacy. IoT devices/sensors are widely used in many fields where users' sensitive private data are collected, transmitted, and processed. For example, a smart home system may handle users' location data, daily hobby data, and even health data. Hence, privacy issues in the IoT must be carefully considered.

DLTs, such as Bitcoin, already provide pseudo-anonymity by using public keys (Bitcoin addresses) as the identifier of users. Yet it remains to tackle the heavy computational cost of blockchain systems. In that direction, Dorry et al. introduced a lightweight scalable blockchain (LSB) [63] for the IoT that can reduce both the computation and bandwidth costs. In a cluster of devices, one node with sufficient resources, referred to as the Overlay Block Manager, is elected to generate or process the transactions related to the cluster members. The authors also propose a time-based lightweight consensus algorithm, in which each Overlay Block Manager must wait for a random time before generating a new block. Similarly to Bitcoin, the anonymity of Overlay Block Managers is ensured by using changeable public keys.

User control policies are also used to protect the privacy of users in the system. In Reference [64], Zyskind et al. proposed a privacy-enforcing distributed computation platform (called Enigma) based on the off-chain distributed hash-table. Compared with traditional blockchain systems, privacy is enhanced by using secure multiparty computation. In Reference [65], by combining the

anonymous blockchain and zero-knowledge proofs, the authors provided a blockchain-based architecture (named ChainAnchor) to deploy IoT devices and sell IoT data in a privacy-preserving fashion.

3.1.4 Trust. Trust is another important issue in the IoT. Although a number of trust management systems exist in literature, a fully distributed solution to meet the requirements of the IoT is still missing [47]. Unsurprisingly, distributed ledger technology is emerging as a promising solution approach to tackle this issue.

By modifying the data structure of Bitcoin's block, the work in Reference [66] proposed a blockchain-based trust management solution for decentralized sensor networks. Due to the tamper-proof property of a blockchain, the behaviour of each node in the network over time can be tracked. This results in a distributed solution for the reliable evaluation of trust without a third party.

The work in Reference [67] established a trust blockchain based on a reputation system, which is used to collect and evaluate past behaviors of peers. In Reference [68], the distributed trust management solution consists of three layers: local layer, shards layer, and global layer. The local layer contains a local private distributed ledger, where all participants access the Internet through the same access point, and the access point bridges the local layer with the shards layer. As a result, the access point should be protected from attacks, failures, and overload. The shards layer is responsible for maintaining the access delegations and reputation scores of a specific subset of devices or users with a permissioned distributed ledger. The global layer is a global, public distributed ledger where each participant is a shards layer entity with a unique identity.

In Reference [70], the authors assume that an IoT device can trust other devices if and only if they are in a zone (i.e., bubbles) of trust. Each communication between devices is regarded as a transaction and must be verified by the blockchain system. Similarly to Reference [63], a single device acts as the manager for each zone. That device is called the master, whereas the other devices are called followers. The master issues *tickets*, which resemble certificates, that include the IDs of the zones, the identities and public addresses of followers, and the signature of the master. Then, a follower can communicate with others after its ticket has been verified via a smart contract in Ethereum.

3.2 DLTs for Identity Management in IoT

In IoT, it is important to assign each device a global unique identifier (GUID), which enables it to connect and communicate effectively and securely with other devices. The identity of things (IDoT) is the cornerstone for applications of IoT. With the maturity of IoT, the number of things connected to the IoT will be out of the capacity of the current identification protocol based on IPv4. Hence, IPv6 is proposed to be a scalable potential solution to this issue. However, both the protocols of IPv4 and IPv6 require a central entity to oversee the arrangement of addresses [13]. Moreover, both the metadata and ownership of devices can be changed and updated in the their lifecycles from design, manufacture, distribution, retail, delivery, deployment, and service to reclaim [71]. Furthermore, traditional centralized identity solutions compromise the privacy of users. A trustworthy, secure, and dynamic identification mechanism for IoT devices should be considered at high priority.

Distributed ledger technologies have the ability to solve these issues in a distributed and secure way. First, DLTs have sufficient addressing capacity to provide a GUID for each device in IoT, noting that there are 2^{160} addresses in Bitcoin theoretically. Also, no centralized authority is needed to manage these addresses. Moreover, since the records in the distributed ledger cannot be tampered with, it can be applied to track and monitor any updates or changes of things securely and efficiently.

To prevent Sybil attacks in mobile ad hoc networks, Bochem et al. [72] proposed an identity system based on the Bitcoin network. The identity proof of a new device is created by sending deposits to a given address. Furthermore, the unique ID in identity proofing is the hash value of the combination of the hash of the block containing the deposit transaction and the index of deposit transaction in that block. As a result, two untrusted nodes can verify the identity proof of each other separately through the blockchain before communicating.

In Reference [73], an identity management system is built on the permissioned blockchain. The group membership of devices is considered as an attribute of identity. In this case, a cryptographic membership authentication scheme designed by Lin et al. [74] can be used to support the group membership.

3.3 DLTs for Machine-to-Machine Transactions

Machine-to-Machine (M2M) communication, which is the foundation of IoT, enables things to share data or collaborate with each other to achieve the same goal without human intervention. As the rapidly emerging application of machine learning in data analysis and decision making, devices or machines could have the ability to make transactions by themselves, perhaps even better than humans in terms of efficiency and accuracy. For example, your smart home could automatically sell redundant power produced by solar panels to the power firm with the highest price. However, the traditional payment methods cannot break away from the involvement of the third party, which could suffer from potential security and privacy issues. The high transaction fee is another important challenge taking into account the large number of machine-to-machine transactions in the IoT.

As a secure peer-to-peer digital currency system, a distributed ledger can form the basis on which devices can make transactions autonomously [82]. Compared with the traditional payment approaches, a DLT offers advantages in terms of security, privacy, costs, efficiency, and flexibility [83]. Most of the current study in this field focuses on how to secure the trading process in M2M. Based on Bitcoin, the work in Reference [84] implements a demo in which a device can pay for electricity autonomously. However, the main drawback of the system is that the transaction fee of Bitcoin is still relatively high compared to the (usually small) value of the transaction. In Reference [75], the proposed blockchain-based IoT data storage system also supports data trading. In the system, the buyer can post the request transaction, which is used to transfer a deposit to the seller. Then the seller can update the access control by constructing a transaction. After both of these two transaction are confirmed by the blockchain, the buyer will get access to the data, and the seller will get the deposit; otherwise, both of them get nothing. As in previous systems, the transaction fee for the block miner is settled in this system. We observe that, in this context, DAG-based and BFT-based ledger systems with little transaction fees seem promising for high volumes of micro-transactions in the IoT.

3.4 DLTs for Traceability and Provenance

Traceability and provenance retain the records of the derivation and the evolution of data over time. Such records can be used for ascertaining the quality of data, reassuring trust on data, detecting errors and abnormal behaviors, and tracking the operations on data by devices. This is useful in applications, such as supply chain [76], smart factory [77], and healthcare [78], where a variate of companies and consumers are involved in the lifecycle of a product.

Large-scale distributed IoT systems, however, make it difficult to achieve traceability and provenance in a secure way. On the one hand, centralized models, which are the de-facto solution to traceability and provenance, cannot accommodate the number of devices expected from this type of system. Moreover, centralized solutions come with single point of failure and the risk of

Table 3. The DLTs Solutions to the Fundamental Security Requirements of IoT

Refs.	Platform	Smart contract	Implementation	Confidentiality	Integrity	Authentication	Access control	Privacy	Trust
[50]	Self-created	×	✓	✓	×	✓	×	✓	✓
[51]	Ethereum	✓	✓	✓	✓	✓	×	×	×
[55, 56]	Self-created	×	✓	✓	×	✓	✓	✓	×
[57]	Self-created	✓	✓	✓	×	✓	✓	✓	×
[58]	Ethereum	✓	✓	✓	×	✓	✓	✓	×
[59]	Ethereum	✓	✓	✓	×	✓	✓	✓	×
[60]	Ethereum	✓	✓	✓	✓	✓	✓	✓	×
[61]	Ethereum	✓	✓	✓	✓	✓	✓	✓	×
[62]	Self-created	×	×	✓	×	✓	✓	✓	×
[63]	Self-created	×	✓	✓	✓	✓	✓	✓	×
[64]	Self-created	×	✓	✓	✓	✓	✓	✓	×
[65]	Self-created	×	×	✓	✓	✓	✓	✓	✓
[66]	Self-created	×	×	✓	✓	✓	✓	✓	✓
[67]	Self-created	×	✓	✓	✓	✓	✓	✓	✓
[68]	Not specific	×	×	✓	✓	✓	✓	✓	✓
[69]	Not specific	×	×	×	×	×	×	×	✓
[70]	Ethereum	✓	✓	✓	✓	✓	✓	✓	✓

physical tampering. An additional challenge is how to cope with the fast data generation from a large number of devices located within heterogeneous networks. In this scenario, recording the workflow information accurately, in timely manner, and with security and privacy guarantees for all involved is an increasing concern.

DLT has the potential to address the above issues. First, the distributed and decentralized nature of DLTs can eliminate the single point of failure. Second, the smart contract can be used to collect the dynamic data accurately, automatically, and in a timely manner. Furthermore, the tamper-poof feature of DLTs ensures that the data are settled in the ledger securely and orderly. Meanwhile, the privacy can be guaranteed by the pseudo-anonymous and cryptographic techniques in the shared ledger. How DLT can be used for traceability and provenance in supply chain will be explain further below in Section 4.3.

3.5 Summary

Security and privacy are critical for IoT with a huge number of devices and sensitive data. Emerging distributed ledger technology is a potential solution to these challenges due to its intrinsic decentralized, trustless, tamper-proof, transparent, and cryptographic security features. In the first part of this section, we have explored how DLTs enhance each aspect of security and privacy in IoT. Undoubtedly, DLTs are paving the way for a secure IoT ecosystem. A summary of previous DLTs solutions to IoT security and privacy issues discussed above is shown in Table 3. Other aspects of IoT security, such as auditability and accountability, can be aided by DLTs [75, 79–81]. The role of DLTs in identity management, M2M transactions, traceability, and provenance in IoT-based applications is highlighted. Table 4 shows which features of DLTs can meet the requirements of IoT.

Approaches integrating IoT with DLT can be divided into two types. Those developing their own IoT distributed ledger system from scratch, and those relying on existing distributed ledger platform (Ethereum, mostly) to accommodate IoT applications. The first approach can provide

Table 4. Mapping DLT Features to IoT Requirements

IoT Requirements	DLT Features
Security and Privacy	1. Distributed and Decentralized 2. Cryptographic Security 3. Tamper-proof 4. Automation & programmability 5. Pseudo-anonymity
Identity management	1. Distributed and Decentralized 2. Cryptographic Security 3. Tamper-proof 4. Automation & programmability
M2M Transactions	1. Distributed and Decentralized 2. Cryptographic Security 3. Tamper-proof 4. Automation & programmability
Traceability and Provenance	1. Distributed and Decentralized 2. Cryptographic Security 3. Tamper-proof 4. Transparency 5. Automation & Programmability

customized functionality for various needs. Yet it imposes a heavy burden on the system designer and does not benefit from the scrutiny that open and popular platforms are typically exposed. The second approach can focus on the requirements of the IoT application, typically by means of coding smart contracts. In this case, the smart contract can be regarded as the interface for IoT applications on the DLT.

4 APPLICATION SCENARIOS USING DLTS

In this section, we explore a number of use-cases of DLTs in specific IoT applications, such as smart home, smart transportation, supply chain management, healthcare, and smart grid.

4.1 Smart Home Using DLTs

A "smart home" is a residence integrated with various smart appliances that work to provide a comfortable and safe place [85]. The sensitive nature of the information handled by a smart home system leads to architectures with a strong focus on security and privacy. For example, Dorri et al. [86–88] have come up with a lightweight blockchain-based architecture for the smart home, which consists of smart homes, one overlay network, and a cloud center for storage. In the proposed architecture, each smart home contains a local blockchain, a local storage and all smart devices located in the home. However, unlike Bitcoin, the blockchain in each smart home is totally controlled by house owner. Moreover, to reduce the associated overheads, the mining process in the blockchain is no longer based on PoW or other puzzles. That is, once a transaction is received, it is treated as a true transaction and automatically put in a block. In this context, a particular kind of device, called *home miner*, is introduced to perform such a mining process. Similarly to Bitcoin, all transactions of the smart home are grouped together and mined by home miner in units of blocks. In addition, the home miner is also used to register and unregister devices by creating or delete the corresponding ledgers of devices. The overlay network is a P2P network

that maintains a public blockchain like Bitcoin. It is responsible for managing the public keys to access data of the smart home. It connects all local smart home and other devices like smart phones or personal computers. Here, a distributed trust mechanism is used to verify blocks and to eliminate the overhead of proof-of-work (PoW). In cloud storage, the data of a user are stored in an identical block. Then the hash of stored data can be used for authentication and identification. The security analysis of this proposed architecture shows that most security requirements could be achieved [87]. The performance evaluation indicates that the overheads incurred by the introduction of blockchain technology are acceptable for the significant security and privacy it offered.

Aung et al. [89] presents a smart-home implementation based on a private Ethereum. This architecture can be regarded as *the smart home* part of the architecture proposed by Dorri et al. Here, the home miner is also used to maintain the private blockchain. The difference is that the access control polices are built on smart contracts of the private blockchain of Ethereum. However, both the security analysis and the performance evaluation are not available.

In Reference [90], the authors propose a distributed ledger system for smart home based on hypergraph. In hypergraph, one hyperedge may connect over two vertexes, and hence, in the system, the vertexes in each hyperedge share the same subchain. Further, since each node may belong to multiple hyperedges at the same time, each node may contains multiple subchains synchronously rather than the entire ledger. As a result, the hypergraph-based ledger can reduce the storage consumption but increase the complexity of consensus, which involves how to maintain lots of subchains synchronously.

4.2 Smart Transport Using DLTs

Smart transport is a notion that aims to offer a secure and efficient transportation system involving vehicles and roadside infrastructure units. Such a system is targeted to offer many services, including finding parking spots, adaptive traffic control, reporting traffic accidents, and making a car insurance claim. Classical solutions to smart transport system are centralized and suffer from the single point of failure problem [91]. As DLTs offers features, such as distribution and decentralization, the integrated application of DLTs and IoT in smart transport systems starts to gain more attention.

For example, in Reference [91], the authors propose a security architecture for smart transport based on a lightweight scalable blockchain (LSB) [63], which was discussed in Section 3.1.3. Due to the limited resource and mobility of vehicles, the Overlay Block Manager (OBM) is introduced to maintain the blockchain. The overlay block manager is a fixed network infrastructure with adequate resources and a constant supply of electricity. As the vehicle shuttles through the city, it will automatically choose the closest OBM to connect via the wireless vehicle interface. In this way, the vehicle can securely communicate with others or backup storage by sending transactions to the blockchain through OBMs. The authors also give some potential applications of this architecture, including insurance, car share services, and software updates. Since the key management scheme plays an important role in securing smart transport, the work in Reference [92] proposed a blockchain-based key management framework. In this framework, the centralized security managers are replaced by the blockchain network, which is used to manage the communication keys of vehicles. It is shown that, with the blockchain, the efficiency and security of key management is improved.

SpeedyChain [93] is a blockchain-based framework for trusted and privacy-preserving data sharing between vehicles, roadside infrastructure units (RSIs), and other participants in the smart city scenario. In the model, the vehicle can post the transaction to the nearest RSI with it signature, whereas the RSIs, which have sufficient resources, are used to maintain the blockchain. In the initial phase, a vehicle is registered with the system based on the location validation of RSI.

The vehicle is authenticated using the public key with an expiration time. After that, the vehicle can upload the collected data by sending transactions to the nearest RSI directly or through other vehicles. On receiving the transaction, the RSI will verify the transaction. If it is successful, then the RSI will append it to a block and broadcast the block over the network.

Trust and privacy are important open issues in smart transport systems. For example, it is still a challenge to report real-time trustable traffic information in an anonymous way. We observe that, in this case, DLT can provide fully decentralized trust among anonymous vehicles and other devices involved [95]. In Reference [96], a blockchain framework with a local dynamic blockchain (LDB) and a main blockchain is used to secure the trusted communication in intelligent vehicle and to meet the efficiency requirement of real-time traffic, the concept of branching of the blockchain is customized for the framework. In References [97, 98], the authors propose a blockcain-based privacy-preserving trust model in vehicular networks, named blockchain-based anonymous reputation system (BARS). BARS consists of three blockchains: The message blockchain is used to store the message sent by vehicles; the certificate blockchain is responsible for recording the issued certificates; and the third blockchain contains all revoked certifications. Furthermore, a reputation evaluation mechanism is proposed based on the behaviors of vehicles, which can be traced on blockchains. The honest and active behaviors will get rewards, whereas misbehaviors will be punished. Similarly, based on a PoW blockchain, the work in Reference [99] provided a reputation system for ensuring the credibility of transferred data in vehicular networks. In Reference [94], a BFT-based blockchain architecture for the privacy-preserving vehicular network is proposed. The blockchain technology is used to trace the malicious nodes and incentivise users to share real-time traffic information anonymously.

Other areas of transport that could benefit from the distributed ledger technologies include auto-payments and car insurance. Smart contracts provide an efficient, secure, and low-cost payment mechanism that can be applied for auto-paying for parking fees, refuelling charges, expressway tolls, and so on. The work in References [100–105] focus on the DLTs-based energy model for secure payments and interactions in the charging process of electric vehicles. Further, due to the traceability and tamper-proof property of distributed ledger technology, it can also be used for auto insurance [106–109]. With smart contracts, an insurance claim can be automatically generated and sent to the distributed ledger with additional information collected by the vehicle. Then, the claim can be confirmed based on the historical records of the vehicle and evidence on the blockchain submitted by other witness vehicles.

4.3 Supply Chain Using DLTs

Today, IoT technology has also been adopted in supply chains. The Internet of Things, GPS chips, sensors, and radio frequency identification technology, can be used to track packages, materials, and containers, solving the problem of monitoring products from the source in real time. The issue of identity authentication in the Internet of Things is particularly essential. One of the advantages of the blockchain is that it provides a practical solution for identity management [110]. Using blockchain technology, the behavior of the performers in the supply chain and when and where they occur are accurately recorded. Supply chain refers to a system of participators, activities, and dynamical information involved in the travel of a product from manufacturer to end customer. At the present stage, such systems are maintained centrally by third parties, which might suffer from numerous security and privacy problems as well as are subject to a trust crisis. In this context, the distributed ledger technology is a promising solution to the application of supply chain by providing tamper-proof and transparent information during production and distribution, which could be trusted by all the involved parties, ranging from manufacturer, distributors, transporters, retailers, deliverer to customers [111].

In Reference [112], Lu and Xu present a private blockchain platform, named OriginChain, to ensure the tamper-proof traceability for supply chain. Due to the limited storage of a block, the traceability certificate is stored in a centralized database, whereas its hash with a small amount of critical or essential data is stored in OriginChain. In addition, as a private platform, OriginChain also provides an on-chain permission management for the access of authorized users. Some smart contracts are deployed to obtain and verify the traceability data through the blockchain. Moreover, to protect privacy, the sensitive data are stored in the centralized database due to the transparency of blockchain.

In Reference [113], the authors present the CoC, a supply management system based on two distributed ledgers, to overcome the performance limitation of transitional distributed ledger systems. CoC consists of two ledgers: one reservation ledger and one data ledger. A two-step block construction method is used to append new blocks to the ledgers. To maintain the public distributed ledgers, CoC introduces a type of participant, named helpers. First, a new reservation block is mined by the helpers who receive the reservation requests of users based on PoW. Then, the data block containing the user's record can be created directly if his/her reservation block is verified. Due to the fact that the second step does not require proof-of-work, the throughput of the system could be significantly improved.

The supply chain system proposed in Reference [114] consists of one public ledger and a set of private ledgers. The data of each private event is stored in the corresponding private ledger, whereas its hash is contained in the public ledger. Similarly, the agricultural supply chain system in Reference [115] is based on double chains: the user information chain and the trading chain.

Traceability and provenance are of great importance to the food supply chain as well. The emergence of the distributed ledger provides a potential solution to these issues [116]. Mao et al. [117], for instance, proposed a blockchain-based credit evaluation system to ensure food safety. The Hyperledger is used as the underlying platform, and deep learning technology is used for the credit evaluation.

SKUchain [118] is a trademarked company that applies blockchain technology to develop solutions for the supply chain. The company is committed to improving trade finance in the supply chain and electronically digitizing it. SKUchain uses smart contract technology, and its system automatically records transactions and automatically processes eligible orders and contracts. This improves the efficiency of trading and financing. SKUchain has developed a range of platforms and tools to address the pain points in the supply chain. The ec3 platform is a management platform designed for the entire supply chain, using the latest blockchain technology. These technologies include popcodes, tracking products, and their component data. Knowledge collaboration technology provides encrypted access control for blockchain users. Currently, SKUchain has cooperated with the Commonwealth Bank of Australia to achieve a blockchain cross-border trade. They shipped a batch of cotton from the United States to China. In this process, the combined application of IoT and blockchain has been realized.

4.4 Healthcare Using DLTs

The healthcare system is responsible for managing huge amounts of electronic health records (EHRs), which contain a highly sensitive and private data from patients. As a result, security and privacy are key. The healthcare industry currently deals with this issue in a centralized way, where the EHRs are maintained by the healthcare providers. This would contradict the patient-centric ideology in modern healthcare. In this context, distributed ledger technology is regarded as a technology that is potentially revolutionary to the entire system of healthcare [119] by distributing the management of health records securely.

With the aid of DLTs, EHRs could be securely stored and shared in the distribution network. For example, in Reference [120], the authors proposed a blockchain-based medical system, namely MeDShare, to provide secure sharing among multiple parties involved. MeDShare consists of four layers: the User layer, the Data Query (DQ) layer, the Data Structuring and Provenance (DSP) layer, and the Database (DB) layer. The types of users are defined in the user layer. The DQ layer contains a querying system and the trigger component, which is used to interact with smart contracts. The DSP layer is mainly used to maintain the blockchain network. Smart contract-based access control is applied in the DSP layer to secure data sharing. The DB layer contains a set of existing databases.

In Reference [121], MedBlock is introduced to secure EMR sharing. In MedBlock, EMRs are stored in the cloud database and can be only accessed by authorized users. The participants in this blockchain system are classfied into four types: clients, endorsers, orderers, and committers. Among them, orderers are used to carry out the modified Byzantine Fault Tolerance consensus algorithm. Moreover, an access control protocol is used to ensure that the sensitive data can only be accessed by the authorized user.

The work in Reference [122] constructs an EHR system by combining two types of blockchains: the private blockchain is used to store the EMRs of patients and the consortium blockchain maintains the indexes of EMRs. The authors also provide a new consensus mechanism, named Proof-of-conformance, which requires the verification of over two thirds of nodes in the blockchain.

DLTs could also enable the aggregation of heterogeneous data from the wearable devices of patients and different healthcare systems [123]. For example, in Reference [124], a blockchain is applied to secure and share health data in a pervasive social network. Also, the work in Reference [125] proposes a smart contract-based private blockchain to secure and manage remote medical devices.

The work in Reference [126] provides a blockchain-based solution for health data sharing. In their proposed framework, the data are collected by users' personal wearable devices and then stored in a centralised cloud database. Meanwhile, the hash of each record is stored in a blockchain. Moreover, the blockchain also acts as the access control interface of the cloud, which ensures the accountability and traceability of data access.

The work in Reference [127] utilizes the blockchain to secure the patient real-time monitoring as well. The authors also provide a Hybrid miner selection approach to address the computation issue of PoW. Similarly, to meet the requirements of security and privacy in a blockchain-based medical system, fundamental signature and authentication issues are addressed in Reference [128] and Reference [129].

4.5 Smart Grid Using DLTs

Electric power is key in our daily life and business activities, and, obviously, the security and availability of the smart grid is critical. The integration of DLTs into smart grids could secure both the grid and energy exchanges. For example, in Reference [130], the authors present a blockchain-based monitoring system (GridMonitoring) to provide transparency and provenance of events in a smart grid. GridMonitoring contains a sovereign blockchain with attached side chains. Each parent block in the sovereign blockchain might contain a set of ordered side blocks, which securely maintain a cluster of contiguous logs.

Considering the potential privacy disclosure in the trading of a smart grid, blockchain is applied into a smart grid to provide privacy-preserving authentication of users based on zero knowledge proof in Reference [131]. In the mining process, the miner is elected according to the accuracy of calculation. More specifically, each node can calculate the average electricity consumption of the data received from other anonymous users. Then, one of them whose electricity consumption is closest to the average is responsible for generating a new block. However, due to the propagation

delay of the network, users might collect data from different users, which would produce different averages. In this situation, several new distinct blocks might exist in the network at the same, which would lead to forking.

In Reference [133], the authors propose a decentralized energy trading system, named PriWatt, based on Bitcoin. Multi-signature based on the Bitcoin script is applied in PriWatt. Moreover, PriWatt strictly requires that the messaging address could be used only once to guarantee the anonymity of users. Hence, PriWatt provides secure and anonymous energy trading transactions with built-in tokens. The work in Reference [134] aims to utilize the consortium blockchain to secure the trading in a smart grid. The proposed consortium blockchain is based on a modified Practical Byzantine Fault Tolerance (PBFT) consensus algorithm and is supplied with inherent energy coins as well. Furthermore, based on a Stackelberg game, the authors present an optimal pricing strategy on the trading blockchain. Similarly, the work in Reference [132] deals with the privacy issue in the smart grid based on the combination of a Bitcoin-like public blockchain and a game platform. However, the authors did not focus on the detail architecture of their blockchain.

4.6 Summary

With the maturity of modern information technologies, such as IoT, big data, communication networks, and intelligent manufacturing, there is an increasing concern for a secure and integrated smart city system that covers the requirements for smart home, transportation, healthcare, and other aspects of intelligent life. Security, privacy, and efficiency have always been at the center of concern in the IoT-enabled smart city. The emerging distributed ledger technology shows solutions to these issues faced by the traditional centralized approaches. For instance, in Reference [135], the authors propose a DLT-based security framework for a smart city. In this framework, the distributed ledger technology is used to guarantee the security and privacy of data in the communication layer. According to the smart city framework in Reference [135], the work in Reference [136] proposed a blockchain-based CitySense system focusing on management of environment data in an urban setting. The work in Reference [137] proposed a distributed blockchain network for a smart city, in which a visible light communication technique, Li-Fi, is used as the medium of communication. To meet with the requirements of trust and audit, the work in Reference [138] proposed a smart contract-based delegation-enabled authorization model for *Stack4Things* [139], which is an IoT-Cloud framework for resource control and management of sensing data in the smart city. Tables 5 and 6 list a set of existing distributed solutions for different aspects of an IoT-enabled smart city.

As shown in Figure 3, the distributed ledger systems are regarded as the underlying network infrastructure for the integration of different components of smart cities. Due to the limited computation and storage of most devices (sensors, mobile, vehicles, etc.) in IoT, some selected devices (for example, the block managers in Reference [63] and the roadside infrastructure units in Reference [93]), should be responsible for the management of local devices and communications with the distributed ledger network. Since there is going to be a vast amount of data generated in IoT, and the storage of current distributed ledger systems are limited, cloud-based storage might be introduced to store the heavy data, such as in the smart home system [86–88] and the healthcare system [121]. In this context, the distributed ledger system provides a secure and privacy-preserving platform for both data exchange and payment in every aspect of the smart city without a centralised controlling intermediary.

5 DISTRIBUTED LEDGER PLATFORMS FOR IOT

In this section, we will report on a range of existing distributed ledger technology-based platforms for IoT. Since a variety of new distributed ledger platforms continues to emerge, rather than being

Table 5. Comparisons of Study Utilizing DLTs for Different Application Scenarios in IoT-enabled Smart City

Refs.	Application Scenario	Platform	Smart Contract	Implementation	Security Analysis	Performance Analysis	Key features
[86], [87], [88]	Smart home	Self-created	×	×	✓	✓	1. Three-tier architecture: smart homes, overlay network and cloud storage. 2. Private chain used for each smart home. 3. One public chain used in overlay network.
[89]	Smart home	Ethereum	✓	×	×	×	1. Private chain based. 2. Smart contract based.
[90]	Smart home	Self-created	×	×	✓	✓	Hypergraph-based structure.
[91]	Smart transport	Self-created	×	×	✓	✓	Based on the LSB [63].
[93]	Smart transport	Self-created	×	×	✓	✓	1. Permissioned blockchain ledger. 2. The roadside infrastructure units are used to support the location-based trust model and to maintain the blockchain.
[96]	Smart transport	Self-created	×	✓	✓	✓	1. Two-blockchain structure: a local one and a main one. 2. Using blockchain branch for efficiency.
[94]	Smart transport	Self-created	×	×	✓	✓	1. A privacy-preserving vehicular announcement protocol. 2. BFT consensus based.
[97], [98]	Smart transport	Self-created	×	×	✓	✓	1.Three-blockchain structure. 2. Roadside units are used to verify the certificate transactions. 3. A Reputation evaluation algorithm.
[99]	Smart transport	Self-created	×	×	×	✓	1.Proof-of-work Consensus based. 2. Selected vehicles act as miners. 3. Reputation based on vehicles' traceable behaviors on blockchain.
[112]	Supply Chain	Self-created	✓	×	×	×	1. Private blockchain platform. 2. Only Hashes of the traceability certificate and small amounts of critical or essential data are stored on the blockchain.
[113]	Supply Chain	Self-created	×	✓	✓	✓	1. Two-step block generation method. 2. Two ledgers system.
[114]	Supply Chain	Self-created	×	✓	×	×	1. One public ledger and some private ledgers. 2. The hashes of private events are stored in the public ledger.
[115]	Supply Chain	Self-created	×	×	×	×	1. Public ledger. 2. Modified proof-of-stake. 3. Double chain architecture.
[117]	Supply Chain	Hyperledger	×	×	×	×	1. Hyperledger-based credit evaluation system. 2. Deep learning based.
[120]	Healthcare	Ethereum	✓	✓	×	✓	1. Four-layer architecture. 2. Smart contract-based access control.
[121]	Healthcare	Self-created	×	×	✓	×	1. Cloud storage. 2. Modified PBFT consensus mechanism.
[122]	Healthcare	Self-created	×	✓	✓	✓	1. Double chain structure: a private blockchain and a consortium blockchain. 2. Proof of conformance consensus mechanism.
[124]	Healthcare	Not specific	×	✓	✓	✓	Pervasive social network based.
[125]	Healthcare	Ethereum	✓	✓	✓	×	1. Private blockchain. 2. Smart contract based. 3. Remote patient monitoring system.
[126]	Healthcare	Hyperledger	×	✓	×	✓	1. Hyperledger-based private blockchain. 2. Data collected by wearable devices. 3. The hash of data is stored in blockchain.

Table 6. Comparisons of Study Utilizing DLTs for Different Application Scenarios in IoT (Continued)

Refs.	Application Scenario	Platform	Smart Contract	Implementation	Security Analysis	Performance Analysis	Key features
[127]	Healthcare	Self-created	×	✓	✓	✓	1. Hybrid miner selection. 2. A secure communication protocol.
[128]	Healthcare	Not specific	×	×	✓	×	Bilinear pairing algorithm for blockchain-based medical system.
[129]	Healthcare	Not specific	×	×	✓	×	Attribute-based signature scheme for blockchain-based medical system.
[130]	Smart Grid	Self-created	✓	×	×	×	A sovereign blockchain with side chains.
[131]	Smart Grid	Self-created	×	×	✓	✓	1. Zero knowledge proof-based privacy preserving in blockchain. 2. Miner is selected according to the average of electricity consumption.
[133]	Smart Grid	Bitcoin	✓	✓	✓	✓	1. Based on Bitcoin. 2. Multi-signature transaction.
[134]	Smart Grid	Self-created	×	×	✓	✓	1. Consortium blockchain. 2. Modified PBFT consensus algorithm.
[132]	Smart Grid	Not specific	×	×	×	×	Bitcoin-like public blockchain.
[135]	Smart City	Not specific	×	×	×	×	Four-layer smart city framework.
[136]	Smart City	Ethereum	×	×	×	×	Based on the framework in [135].
[137]	Smart City	Not specific	×	×	×	×	Secure visible light communication for blockchain-based smart city.
[138]	Smart City	Ethereum	✓	✓	×	✓	1. Smart contract based. 2. An authorization and delegation model for an IoT-Cloud framework (i.e., Stack4Things [139]) in smart city.

exhaustive, we list some typical platforms that meet the requirements of specific IoT domains (see Figure 4).

Bitcoin [21] is the first and the most valuable cryptocurrency in the world. As the first blockchain system, Bitcoin provides an efficient, cheap, and secure payment system. Bitcoin supports the smart contract by using a low-level programming language, the *Bitcoin Script*. *Bitcoin Script* is a stack-based language with a list of useful instructions, such as hash computation and signature verification. The *Bitecoin Script* is intentionally not Turing-complete with no loops, since an infinite loop in transactions could exhaust the hash computation power of the entire network. Besides, the Bitcoin Script is not user friendly and is considered somewhat esoteric. Moreover, the confirmation of transactions in Bitcoin has an average of 10 minutes, which is not acceptable for real-time communications. Thus, it is mainly used as a payment system and difficult to be extended to other IoT applications. However, as a pioneer of distributed ledger systems, the data structure and consensus algorithm used in Bitcoin offer references for many DLTs-based IoT solutions, such as in References [63, 86–88, 91, 133].

Ethereum [44] is the first Turing-complete decentralized smart contract system. With a built-in programming language (*Solidity*), Ethereum serves as a virtual machine (EVM) for varieties of decentralized applications, including but not limited to cryptocurrencies. *Solidity* is a C# like object-oriented language, which is more friendly than Bitcoin Script. The transaction fee in Bitcoin depends on transaction size, whereas in Ethereum, it depends on how much work should be spent for the transaction or contract. In Ethereum, the transaction fee is defined as *gas cost*, where gas

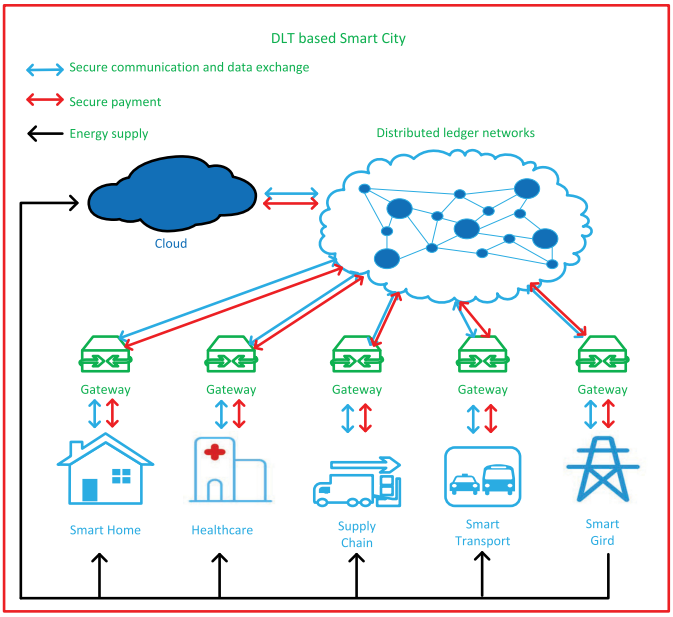

Fig. 3. The DLT-based smart city.

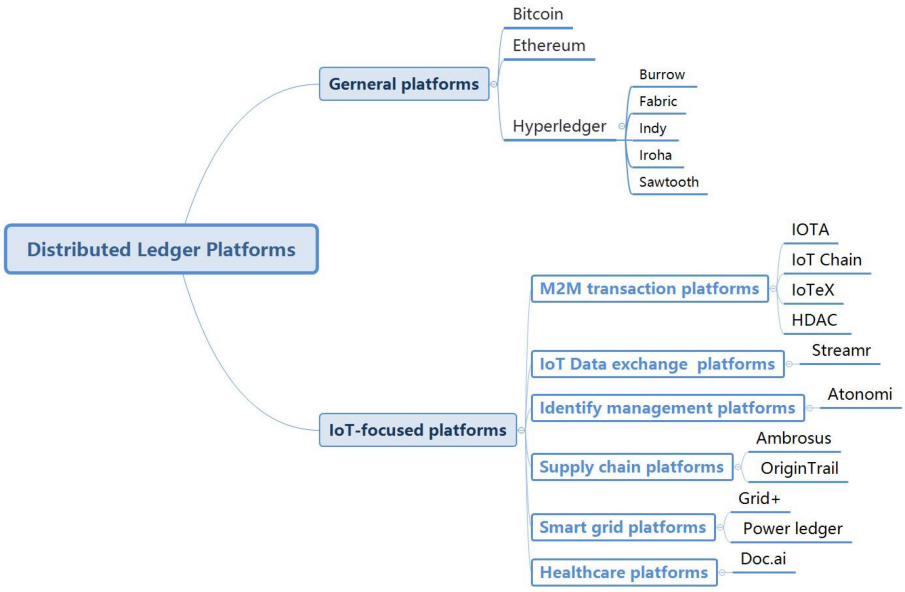

Fig. 4. Some distributed ledger platforms for IoT and their features.

is a unit that is used to measure the computational effort required to execute an operation. For each contract submitted to the network, the transaction fee must be prepaid. Then the contract would be killed if the actual transaction fee exceeds the prepaid value, which stays out of the endless loop. As a result, compared with Bitcoin, Ethereum provides a better deterministic and terminable platform for the smart contract. Benefiting from a mature development community,

Ethereum contains the largest number (over 2,000) of decentralized applications (Dapps) [150]. By utilizing the smart contract, lots of IoT applications can also be implemented on Ethereum (see Tables 3, 5, and 6).

Hyperledger fabric [151] is the main member of the open-source blockchain framework family, *Hyperledger* [34], hosted by the Linux Foundation. It is a PBFT-based permissioned blockchain platform without built-in cryptocurrency. The smart contract (namely *chaincode* in Hyperledger fabric) can be written in Go or node.js. To avoid endless loops, a preset timer is introduced into the chaincode to be capable of termination. As a permissioned blockchain, users must be authorized before performing operations in Hyperledger. Hyperledger fabric is specially formulated for use in commercial applications. Combined with the IBM Watson IoT Platform [152], the Hyperledger fabric can provide enterprise-wide solutions for IoT applications, such as management of IoT devices and data, supply chain, and smart vehicles. Currently, other distributed ledgers in the Hyperledger family are Burrow [153], Indy [154], Iroha [155], and Sawtooth [156]. Among them, Burrow is used to support the EVM of Ethereum, Indy is for the trust privacy of decentralized identity, Iroha focuses on mobile applications, and Sawtooth is a proof of elapsed time-based framework for asset-ownership-like records. Hyperledger also provides frameworks for specific domains. For example, the Hyperledger Grid [157] is a specific framework that focuses on supply chain.

IOTA [22] is one of the first distributed ledger platforms that mainly concentrates on the M2M micropayment in IoT. Unlike Bitcoin and Ethereum, the underlying data structure of IOTA is a directed acyclic graph (the tangle) instead of blockchain. Rather than transactions generated by users being packaged into blocks by the miners in Bitcoin, every user in IOTA must act as miners as well. More specifically, every new transaction must randomly approve two previously unconfirmed transactions. As a result, the network throughput in IOTA scales linearly with the number of transactions being submitted to the network. Notably, there are no transaction fees and transaction throughput limitations, which is suitable for M2M micropayments in IoT. However, the real-time throughout of IOTA is only seven transactions per second at present [158], because there are still few users in IOTA. Besides offering the micropayment services, IOTA also aims to enable the security and privacy of massive IoT data transferred through the tangle.

IoT Chain [159] aims to provide a lightweight system for addressing the security and scalability issues for IoT devices. It combines the PBFT consensus algorithm with the underlying IOTA-like directed acyclic graph, which makes the platform blazing fast. There are two kinds of transactions in IoT Chain: zero-value transactions and value transactions. There is no transaction fee for zero-value transactions, which might refer to an interconnection between devices. However, similarly to Bitcoin, a transaction fee is required for value transactions, which based on the size of the transaction.

IoTeX [160] is a blockchain-in-blockchain platform for M2M transactions with particular concerns for privacy. The blockchain-in-blockchain architecture consists of one public rootchain and some subchains. A subchain can be a public or private blockchain that manages a set of connected IoT devices. The rootchain is used to manage other independent subchains and also supports a non-Turing-complete language. Subchains can communicate with each other by generating cross-blockchain transactions to the rootchain. Due to the limited size of a block in the rootchain, a transaction fee is charged by the cross-blockchain transaction submitted by the subchain. The consensus algorithm used for rootchain is Roll-DPoS, which is an integration of PBFT, delegated proof-of-stake (DPoS) [161], and verifiable random functions [162]. It is designed so that a new block is appended to the rootchain every 3s.

HDAC [163] is a multichain platform that focuses on the M2M transactions and IoT contract services. The consensus algorithm used in HDAC is an energy-saving modification of proof-of-work, namely ePoW, which is based on the block window mechanism. For a given period (i.e.,

the block window), each node can succeed in mining only once. Moreover, the block window can be dynamically changed, which depends on the number of network nodes, mined blocks, and unmined blocks. In a nutshell, ePoW can reduce huge energy wastage by replacing the greedy computation competition in mining with a more equitable mechanism. The IoT contracts in HDAC are smart contracts that are specifically used for the interactions between user and device or device and device. Considering the fact that many IoT devices are developed in C, the script of IoT contract has a syntax similar to C.

Streamr [164] is an Ethereum-based open source off-chain network for trading of real-time data. It provides a global data marketplace that anyone can sell or purchase data with its in-built tokens (DATAcoins). With built-in data-driven smart contracts, connected devices also can autonomously sell their own data and buy the data they want in real time. Each piece of data is incorporated into an event, with its timestamp, origin, content type, encryption type, and other metadata for the event. Events are transferred on a P2P network (Streamr network) through a publish/subscribe mechanism and can be stored in the decentralized network. Since the throughout is in proportion to the number of nodes in a P2P network, it is said that Streamr can handle over millions of events per second. As the consensus is based on the underlying Ethereum, a transaction fee is required not by Streamr but by miners in Ethereum.

Atonomi [165] aims to built a trustworthy and secure IoT ecosystem by providing trusted immutable identity management services. The Atonomi network is built on the underlying Ethereum framework. Atonomi provides a software development kit to original equipment manufacturers to build in the mechanisms of trust by embedding registration and reputation services. Then, registration and reputation services of devices can be activated by the registration smart contract when it is executed. Then, the devices can be traced and audited in its entire lifecycle with smart contracts.

Ambrosus [166] is a distributed IoT platform for synchronizing, sharing, and securing information in supply chains. Ambrosus is an Ethereum-based distributed network, where smart contracts are used to ensure the transparency and immutability of supply chain data. The Ambrosus network supports a variety of IoT devices and sensors to track goods throughout the entire supply chain. All collected data of an item transmitted to Ambrosus must be formatted and bonded to an Amber token. The Amber balance of the item can be split or merged along with the travel, which forms an entire history of the item. Moreover, the underlying InterPlanetary File System (IPFS) is used to store the collected massive data in a transparent and unalterable manner.

OriginTrail [167] is another supply chain data sharing platform based on blockchain. OriginTrail is a decentralized network that consists of three layers: a blockchain layer, a data layer, and a network layer. Ethereum is used as the underlying blockchain platform for the current test version of OriginTrail. The hashes of all records in the OriginTrail mainnet are stored in the underlying blockchain platform to ensure the integrity of data. In data layer, a decentralized graph database is used to store the formatted supply chain data. The network layer is responsible for maintaining the accessibility of the data.

Grid+ [168] is an Ethereum blockchain solution for energy markets. It relies on a smart energy agent, which is used to pay electricity bills in real time. All agents must be registered with a serial number and an Ethereum address by the registry smart contract of Grid+. Then, the agent can automatically purchase energy through the Grid+ system. The smart agent can manage IoT devices in the home. With a built-in lightweight Ethereum wallet, smart agents can be used to generate transactions on behalf of other devices.

Power ledger is an energy trading platform based on smart contracts. It consists of two blockchain layers: a public blockchain and a consortium blockchain. The public blockchain is built on smart contracts of Ethereum, and the consortium blockchain is based on a private chain, called

Table 7. Comparison of Distributed Ledger Platforms for the Application of IoT

Platform	Data Structure	Public Ledger	Performance (TPS)	Transaction Fee	Smart Contracts	Currency	Consensus	Launch Time
Bitcoin	Blockchain	✓	7	✓	✓	✓	Proof-of-work	Jan 2009
Ethereum	Blockchain	✓	15	✓	✓	✓	Switching from proof-of-work to proof-of-stake	Jul 2015
Hyperledger fabric	Blockchain	×	3,500	×	✓	×	PBFT	Jul 2017
IOTA	DAG	✓	1,000 (unlimited theoretically)	×	×	✓	Tangle	Jul 2017
IoT Chain	DAG	✓	Unknown	✓	✓	✓	Combination of PBFT and Tangle	Under test
IoTeX	Multiple blockchains	✓	Unknown	✓	✓	✓	Roll-DPOS	Under test
HDAC	Multiple blockchains	✓	160	✓	✓	✓	ePoW Trust-based	May 2018
Streamr	Blockchain	✓	Unknown	✓	✓	✓	Based on Ethereum	May 2018
Atonomi	Blockchain	✓	Unknown	✓	✓	✓	Based on Ethereum	Dec 2017
Ambrosus	Blockchain	✓	Unknown	✓	✓	✓	Based on Ethereum	Jul 2018
OriginTrail	Blockchain	✓	Unknown	✓	✓	✓	Based on Ethereum	Under test
Grid+	Blockchain	✓	Unknown	✓	✓	✓	Based on Ethereum	Under test
Power Ledger	Double blockchains	✓	Unknown	✓	✓	✓	Based on Ethereum and EcoChain	Under test
Doc.ai	Blockchain	✓	Unknown	✓	✓	✓	Based on Ethereum and EcoChain	Aug 2017

EcoChain [170]. Two types of tokens are implemented on the two blockchains by following the ERC20 standard of Ethereum, respectively. The application layer, namely FuseBox, is designed to provide a trustless and low-cost energy trading by utilizing these tokens.

Doc.ai [171] mainly aims to provide insights of medical research, integrating blockchain technology and artificial intelligence. It utilizes Ethereum as the underlying blockchain framework, which is public for anyone to contribute his/her health data in an anonymous and encrypted way. Then neural network models are trained with collected data, which could provide a custom-tailored treatment for patients and support new results from medical research.

A comprehensive comparison of reviewed platforms can be found in Table 7. As one can see, most current DLT-IoT applications are public ledgers and are based on the popular smart contract platform—Ethereum. On one hand, Ethereum is an open public blockchain where anyone can join freely. However, it provides a mature smart contract framework for the convenient interactions with other on-chain or off-chain platforms. Moreover, Ethereum proposed an ERC-20 standard for implementing new tokens within the Ethereum ecosystem, which has become the first choice of the initial coin offering (ICO) of new crowdfunding companies.

6 OPEN ISSUES AND CHALLENGES

There are a range of open issues and challenges with the application of DLTs to IoT.

- *Security and Privacy of DLTs.* While DLTs aim to address security and privacy issues in IoT transactions and tracking, the security and privacy of DLTs themselves need to be maintained. Bitcoin and Ethereum and DLTs in general are not unhackable.[2] Public

[2]For example, see https://thenextweb.com/hardfork/2018/05/25/you-could-profitably-exploit-a-2-billion-cryptocurrency-with-as-little-as-1-5-million-researcher-shows/, https://coincentral.com/blockchain-hacks/.

blockchains could expose data so that in some systems, private data are stored off-chain or in secured parts of blocks. The idea of millions of copies of a ledger in a distributed system can be notionally uncomfortable, with the confidence on encryption to enable such data to be tamper-proof and unreadable to unauthorized users. Also, execution of (especially Turing-complete) programs as smart contracts can be dangerous. There is emerging work on the need for testing tools for detecting the security vulnerabilities in smart contracts to increase the trustworthiness and safety of smart contracts [140]. The larger increase in the use of DLTs implies a greater need for them to be secured. The notion of multiple DLTs, each for a different purpose, or fragmenting, might be useful in that when one DLT is compromised, there is potential isolation from effects on other DLTs.

- *Interfacing DLTs with the physical systems.* DLTs exist in the cyber world while IoT devices and actuators work in the physical work. While events in the physical world can be sensed to trigger updates in the ledger or execution of smart contracts, the issue of the gap between the physical world and cyber world remains; we want to ensure that this gap is not taken advantage of and events recorded in the ledger do correspond to real-world events. Electronic tagging technologies such as tamper-proof RFID can be employed,[3] but there is a need to then actually tag the physical world items (e.g., from objects to fish [172]) and ensure that tagging does not affect the item being tagged. Veracity Protocol [173] aims to "fingerprint" items by extracting unique material characteristics of items using computer vision inspection, e.g., to detect counterfeit money.[4] IBM's Crypto Anchor Verifier [174] scans for optical characteristics to uniquely identify physical items via optical patterns, applicable to diamonds, wine, and other substances. Apart from identifying physical items, there is a need to identify events that actually take place when they are recorded in a blockchain—proof-of-events technology might be required. It remains a challenge (perhaps application-specific) to ensure that this gap between the physical and cyber worlds is not exploited, and cost-effective technologies to identify physical items may depend on the nature and form of the physical item itself.
- *Expanding system requirements and scalability.* As DLTs increase in size monotonically, storage, energy, and associated costs will increase for their continued operation, with implications on throughput and performance. To reduce the Bitcoin block time, recent work has begun to look at subchains [149].[5] While consensus algorithms can be developed to reduce time-to-verification (such as proof-of-stake and pre-selecting validators instead of proof-of-work),[6] there is a need to consider such system requirements, projecting into the future in any deployment of a DLT solution. Indeed, further studies on performance issues with DLTs can be done.
- *Multi-chain world.* It can be seen from our analysis above that there is a large range of DLT platforms, and new ones will emerge, some tailored to a particular use cases. Also, some DLTs will be public, some private, some hybrid (private and public), and some consortium based [175]. A complex workflow involving medical records, payments, delivery of products, and knowledge of the home environment could involve multiple DLT systems, each managing a particular task in the workflow. Hence, there is an emerging case for multi-DLT operations and systems, and their interoperability, from conversion of cryptocurrencies from one currency to another to how data from one DLT can be used by another DLT.

[3]For example, see https://www.identiv.com/products/rfid-nfc-inlays/rfid-labels/.
[4]https://www.startuprad.io/blog/the-app-of-veracity-protocol-can-spot-counterfeit-czech-crowns-with-100-accuracy/.
[5]http://community.chainblx.com/t/scalability-of-blockchain-chainblx/211.
[6]See also the recent proposal on Casper [41].

Work on blockchain standards is progressing, e.g., the ISO/TC 307 [176] and IEEE supported efforts with standards activities for blockchain in IoT, in agriculture, in electric power infrastructure, and in autonomous vehicles [177]. The issue is, however, not just of standards, since DLTs can cross application domains and industries, so that standards could be difficult to arrive at. Another perspective and perhaps a potential solution is the market-place of blockchains, where different blockchains compete for users, and users might record transactions on multiple blockchains and employ multiple blockchain services in their applications. Also, for DLTs to be effective, they must exist in scale; the picture of entangled large chains calls for research in cross-relating chains and maintaining inter-chain consistency, e.g., hashing is typically done at the level of blocks in a blockchain but perhaps can also be done at the level of subchains. The recently proposed Blocknet protocol [178] aims to provide a means for nodes on different blockchains to communicate, enabling inter-chain or multi-chain applications. Users will also need convenient interfaces to multi-chains.

- *IoT, edge computing, fog computing, and DLTs.* The case for the use of edge computing and fog computing resources, rather than just centralised cloud resources, for IoT devices has been well argued.[7] With such decentralization of cloud resources, such as the mobile cloud or cloudlets model [144], there needs to be a mechanism for decentralised payments and enforcing trust in open systems of peers; for example, when a device that wants the resources of another device is faced with the issue of whether to trust the other device enough to pay first before having the job done, and the other device has the dilemma of whether to do the job first and get paid later or to insist on payment first. DLTs have been proposed as a solution for peers to perform secure payments (via smart contracts) once jobs have been completed [147]. Indeed, DLTs can be used to record rewards for work done via a token-based economy, e.g., credits for performing computation tasks for others. However, DLT-based trust for such peer-to-peer collaborations is still to be proven on a large scale.
- *Quantum computing.* Developments in quantum computing have raised a number of questions regarding the security of blockchain technologies that relies on secure cryptographic protocols and resource-intensive mining. For example, the work in Reference [142] showed that a quantum computer could be used to break the elliptic curve signature scheme used by Bitcoin by the year 2027 and that a quantum computer could be fast enough to solve the blockchain proof-of-work task to allow one to control the blockchain if a quantum computer with enough qubits could be developed (optimistically estimated to be in 10 years' time). The work in Reference [141] also noted how the quantum computing could challenge proof-of-work schemes.

  For post-quantum security, lattice-based cryptographic schemes have been proposed as a promising approach [146]. There are challenges to energy efficient and computationally efficient implementations in IoT devices of such schemes, which are being addressed, e.g., using the chip developed in Reference [143] and the work in Reference [145] on a FPGA implementation of a key encapsulation mechanism in lattice-based cryptography. Also, a conceptual proposal in Reference [148] uses quantum entanglement in time to provide tamper-proofing, basically employing the idea of quantum networks for a blockchain; while it is argued that all the subcomponents of this scheme have been experimentally realized, a real implementation of the scheme remains. Hence, quantum computing may be a threat to blockchain security but could be a solution to yield a new class of quantum blockchains.

[7]For example, see https://www.i-scoop.eu/internet-of-things-guide/edge-computing-iot/.

7 CONCLUSION

This article has provided an overview of DLTs and their applications in the Internet of Things. We have also outlined a range of issues that need to be addressed in their use and application, ranging from bridging the physical and cyber worlds cost-effectively to preparing for a post-quantum IoT world.

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